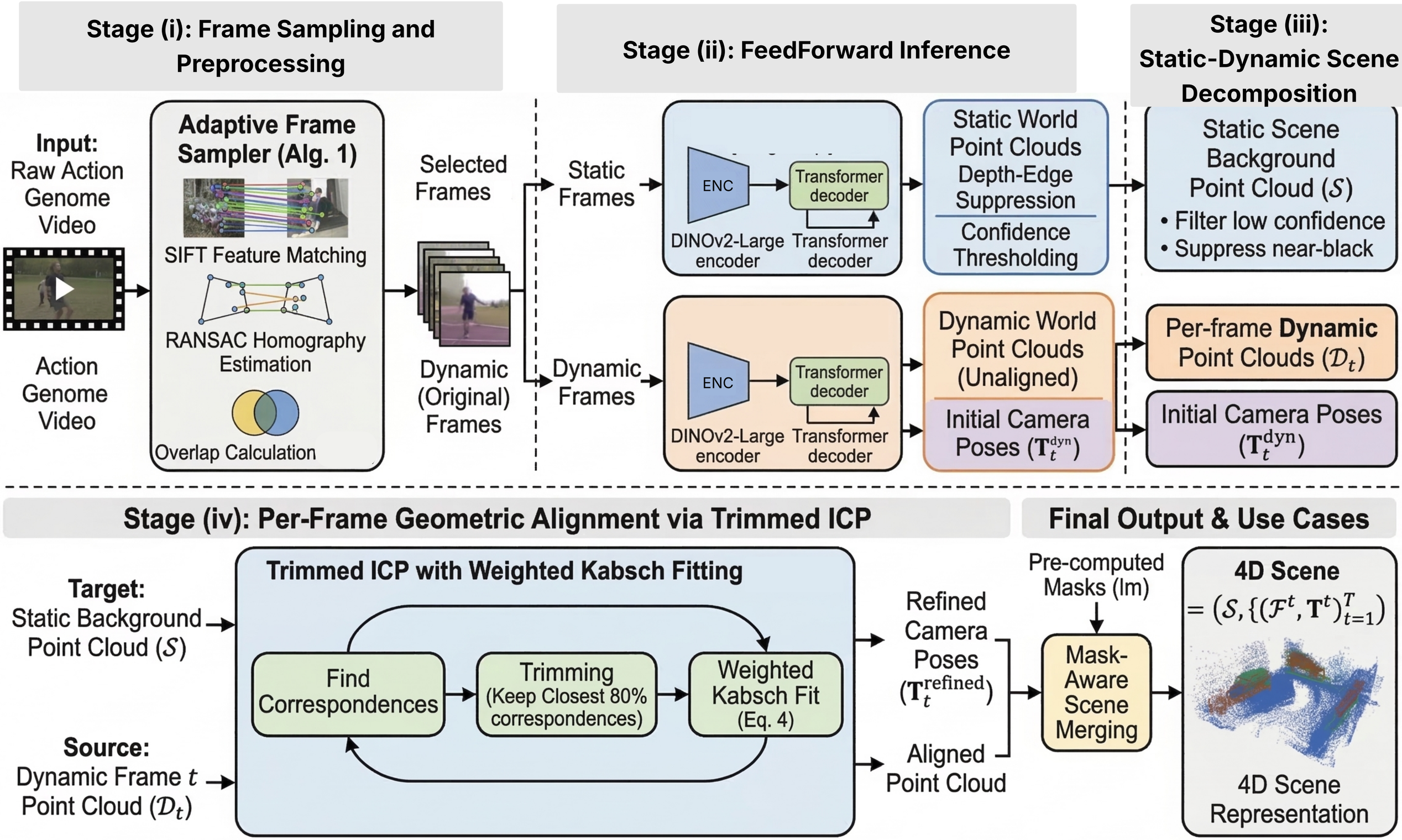
Stage (i): Frame Sampling and Preprocessing
Stage (ii): FeedForward Inference
Stage (iii): Static-Dynamic Scene Decomposition
Input: Raw Action Genome Video
Action Genome Video
Adaptive Frame Sampler (Alg. 1)
SIFT Feature Matching
RANSAC Homography Estimation
Overlap Calculation
Selected Frames
Dynamic (Original) Frames
Static Frames
Dynamic Frames
ENC
Transformer decoder
DINOv2-Large encoder
Transformer decoder
Static World Point Clouds Depth-Edge Suppression
Confidence Thresholding
Dynamic World Point Clouds (Unaligned)
Initial Camera Poses ($\mathbf{T}_t^{\mathrm{dyn}}$)
Static Scene Background Point Cloud ($\mathcal{S}$)
• Filter low confidence
• Suppress near-black
Per-frame Dynamic Point Clouds ($\mathcal{D}_t$)
Initial Camera Poses ($\mathbf{T}_t^{\mathrm{dyn}}$)
Stage (iv): Per-Frame Geometric Alignment via Trimmed ICP
Final Output & Use Cases
Target: Static Background Point Cloud ($\mathcal{S}$)
Source: Dynamic Frame t Point Cloud ($\mathcal{D}_t$)
Trimmed ICP with Weighted Kabsch Fitting
Find Correspondences
Trimming (Keep Closest 80% correspondences)
Weighted Kabsch Fit (Eq. 4)
Refined Camera Poses ($\mathbf{T}_t^{\mathrm{refined}}$)
Aligned Point Cloud
Pre-computed Masks (Im)
Mask-Aware Scene Merging
4D Scene $= (\mathcal{S}, \{(\mathcal{F}^t, \mathbf{T}^t)_{t=1}^T)$
4D Scene Representation